$$\text{Speed} = \sqrt{(6t)^2 + 5^2} = \sqrt{36t^2 + 25}$$

Now the velocity at t = 3sec is given by

$$\vec{v} = 6(3)\hat{i} + 5\hat{j} = 18\hat{i} + 5\hat{j}.$$

and speed at t = 3 sec, is given by

$$\text{speed} = \sqrt{349}\ ms^{-1}$$

2. A gun is fired from a place which is at distance 1.2 km from a hill. The echo of the sound is heard back at the same place of firing after 8 second. Find the speed of sound.

Solution:

The echo will be heard when the sound reaches back at the place of firing. So, the total distance travelled by sound is 2 × 1.2 km = 2.4 km = 2400 m.

$$\text{speed} = \frac{2400\ \text{m}}{8s} = 300\ \text{m s}^{-1}$$

3. A train 100 m long is moving with a speed of 60 kmh^{-1}. In how many seconds will it cross a bridge of 1 km long?

Solution:

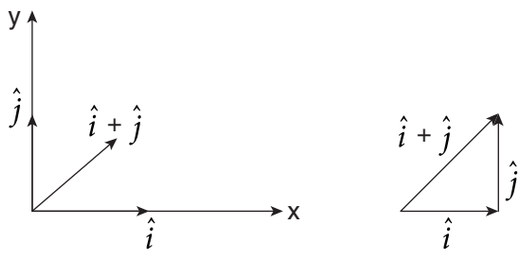
Total distance to be covered = 1 km + 100 m = 1100 m (including both bridge and time)

$$\text{Then, Speed} = 60\ \text{km h}^{-1} = 60 \times \frac{5}{18}\ \text{m s}^{-1} = \frac{150}{9}\ ms^{-1}$$

Then, Time taken to cover this distance $= \dfrac{1100}{150/9}\ \text{s} = 66\ \text{s}$

4. Draw the resultant direction of the two unit vectors $\hat{i}$ and $\hat{j}$. Use a 2-dimensional Cartesian co-ordinate system. Is $\hat{i} + \hat{j}$ a unit vector?

By using the triangular law of addition $\hat{i} + \hat{j}$ as shown in the following figure,



The definition of unit vector is $\hat{A}.\hat{A} = 1$

But here,

$$(\hat{i} + \hat{j}).(\hat{i} + \hat{j}) = \hat{i}.\hat{i} + \hat{i}.\hat{j} + \hat{j}.\hat{i} + \hat{j}.\hat{j}$$
$$= 1 + 0 + 0 + 1 = 2$$

So, $\hat{i} + \hat{j}$ is not a unit vector.

To make any vector to a unit vector, must divide the vector by its magnitude,

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

The norm of the vector $\hat{i} + \hat{j} = \sqrt{2}$.

Hence, the unit vector is $\dfrac{\hat{i} + \hat{j}}{\sqrt{2}}$

5. A swimmer moves across the Cauvery river of 750 m wide. The velocity of the swimmer relative to water ($\vec{v}_{sw}$) is 1.5 m s^{-1} and directed perpendicular to the water current. The velocity of water relative to the bank ($\vec{v}_{wb}$) is 1 m s^{-1}. Calculate the
   (a) velocity of the swimmer with respect to the bank of the river ($\vec{v}_{sb}$).
   (b) time taken by the swimmer to cross the Cauvery river.

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